

Machine Learning

Maths Problem Sets 1–4, with Solutions

Computations, derivations and short proofs for Lectures 2–4. No essay questions.

How to use these sets

These four sets are *computation and derivation only*: every question asks you to calculate a number, derive a formula, or prove a short result. They follow the lectures: Set 1 is decisions under a known distribution (Lecture 2, first half), Set 2 is learning from data (Lecture 2, second half), Set 3 is generalisation (Lecture 3), Set 4 is discriminative and generative classification (Lecture 4).

Each problem opens with a blue box holding the shared setup, then a yellow box listing the formulas you need with a pointer [\[one-read §x\]](#) to where they are explained in the one-read document. Each sub-question sits in its own bar, with its full solution directly underneath. Difficulty: ★ direct application, ★★ several steps, ★★★ derivation or proof. Problems marked *looking ahead* use a derivative that later lectures on optimisation will need; they are good practice now.

All logarithms are natural ($\ln$) unless written $\log_2$. Where a normal-distribution value is needed, Φ denotes the standard normal CDF and the values needed are given.

Contents

1	Set 1: Decisions under a known distribution	2
1.1	Problem 1.1: Three actions, three classes	2
1.2	Problem 1.2: The pinball loss and a uniform outcome	3
1.3	Problem 1.3: Every summary is optimal for its own loss	4
1.4	Problem 1.4: The reject option in general, and what review is worth	4
1.5	Problem 1.5: Bayes error of two Gaussian classes	5
2	Set 2: Learning from data	6
2.1	Problem 2.1: KNN by hand, and leave-one-out	6
2.2	Problem 2.2: Which distance?	7
2.3	Problem 2.3: Softmax, cross-entropy and their derivatives	7
2.4	Problem 2.4: Entropy, KL divergence and mutual information	8
2.5	Problem 2.5: ERM in closed form	9
3	Set 3: Generalisation	10
3.1	Problem 3.1: Bias and variance of a shrinkage estimator	10
3.2	Problem 3.2: Bias and variance of KNN regression	10
3.3	Problem 3.3: How precise is a test score?	11
3.4	Problem 3.4: Selection noise	12
3.5	Problem 3.5: L_1 and L_2 in numbers	12
3.6	Problem 3.6: Early stopping and patience	13
3.7	Problem 3.7: Leakage through a scaler	14

3.8	Problem 3.8: Precision depends on the base rate	14
4	Set 4: Discriminative and generative classification	15
4.1	Problem 4.1: Logistic regression by hand	15
4.2	Problem 4.2: Maximum likelihood for Gaussian classes	15
4.3	Problem 4.3: LDA and QDA disagree	16
4.4	Problem 4.4: Gaussian Naive Bayes, and why variances need smoothing	17
4.5	Problem 4.5: Multinomial Naive Bayes for spam	18
4.6	Problem 4.6: Prior shift in odds form	19
4.7	Problem 4.7: Confident but unfamiliar	19

1 Set 1: Decisions under a known distribution

1.1 Problem 1.1: Three actions, three classes

Problem 1.1 (★)

An intrusion detector gives each flow posterior probabilities for three classes: *normal*, *minor attack*, *major attack*. Three actions are available: *ignore*, *monitor*, *block*. The cost $c(a, y)$ of action a when the truth is y is:

	normal	minor	major
ignore	0	5	20
monitor	1	0	8
block	4	2	0

Tools you need

Conditional risk $\mathbb{E}[\ell_{\text{dec}}(Y, a) \mid X = x] = \sum_y c(a, y) P(Y = y \mid x)$ and the Bayes rule: take the action with the smallest conditional risk [one-read §2.2].

(a) For posteriors $(0.5, 0.3, 0.2)$, compute the conditional risk of each action and the optimal action.

Weight each row of the table by the posteriors and add:

$$\text{ignore} \quad 0(0.5) + 5(0.3) + 20(0.2) = 1.5 + 4 = 5.5$$

$$\text{monitor} \quad 1(0.5) + 0(0.3) + 8(0.2) = 0.5 + 1.6 = 2.1$$

$$\text{block} \quad 4(0.5) + 2(0.3) + 0(0.2) = 2 + 0.6 = 2.6$$

The smallest is 2.1: **monitor**. Notice that the most probable class is “normal”, yet doing nothing is by far the worst choice, because a 20% chance of a major attack carries a cost of 20.

(b) Now let the posteriors be $(0.8 - q, 0.2, q)$ with $0 \leq q \leq 0.8$. Write the three conditional risks as functions of q and find the optimal action for every q .

$$\text{ignore} \quad R_I(q) = 0 + 5(0.2) + 20q = 1 + 20q$$

$$\text{monitor} \quad R_M(q) = (0.8 - q) + 0 + 8q = 0.8 + 7q$$

$$\text{block} \quad R_B(q) = 4(0.8 - q) + 2(0.2) + 0 = 3.6 - 4q$$

Ignore versus monitor: $R_I - R_M = 0.2 + 13q > 0$ for every $q \geq 0$, so ignoring is never optimal.
Monitor versus block: $0.8 + 7q < 3.6 - 4q \iff 11q < 2.8 \iff q < 2.8/11 \approx 0.2545$.

monitor if $q < 0.2545$, block if $q > 0.2545$ (either at equality).

Check at $q = 0.3$: $R_M = 2.9$, $R_B = 2.4$, so block. ✓

(c) Suppose the actions are the classes themselves and $c(a, y) = \mathbf{1}(a \neq y)$. Show that the Bayes rule chooses the most probable class.

The conditional risk of answering class a is

$$\mathbb{E}[\mathbf{1}(Y \neq a) | x] = \sum_y \mathbf{1}(y \neq a) P(Y = y | x) = \sum_{y \neq a} P(Y = y | x) = 1 - P(Y = a | x).$$

Minimising $1 - P(Y = a | x)$ over a is the same as maximising $P(Y = a | x)$, so $f^*(x) = \arg \max_a P(Y = a | x)$. ■

1.2 Problem 1.2: The pinball loss and a uniform outcome

Problem 1.2 (★★)

For a fixed input x , the output is uniform: $Y | x \sim \text{Uniform}[2, 10]$, density $\frac{1}{8}$ on $[2, 10]$. The asymmetric (“pinball”) loss charges $c_u = 3$ per unit of under-prediction and $c_o = 1$ per unit of over-prediction:

$$\ell(y, a) = \begin{cases} 3(y - a), & y > a, \\ 1(a - y), & y \leq a. \end{cases}$$

Tools you need

Squared loss $\rightarrow$ conditional mean, absolute loss $\rightarrow$ conditional median, pinball loss $\rightarrow$ quantile at level $c_u/(c_u + c_o)$ [one-read §2.2].

(a) Give the optimal prediction under squared loss and under absolute loss.

Squared loss: the conditional mean, $(2 + 10)/2 = 6$. Absolute loss: the median, the point with half the mass on each side, also 6. For a symmetric distribution the two coincide.

(b) For $2 \leq a \leq 10$, show that $\mathbb{E}[(Y - a)_+] = (10 - a)^2/16$ and $\mathbb{E}[(a - Y)_+] = (a - 2)^2/16$, where $u_+ = \max(u, 0)$.

$$\begin{aligned} \mathbb{E}[(Y - a)_+] &= \int_a^{10} (y - a) \frac{1}{8} dy = \frac{1}{8} \left[\frac{(y-a)^2}{2} \right]_a^{10} = \frac{(10-a)^2}{16}, \\ \mathbb{E}[(a - Y)_+] &= \int_2^a (a - y) \frac{1}{8} dy = \frac{1}{8} \left[-\frac{(a-y)^2}{2} \right]_2^a = \frac{(a-2)^2}{16}. \end{aligned}$$

(c) Minimise the expected pinball loss directly and check that you get the quantile predicted by the general formula.

The expected loss is $L(a) = 3 \mathbb{E}[(Y - a)_+] + 1 \mathbb{E}[(a - Y)_+] = \frac{3(10-a)^2 + (a-2)^2}{16}$. Then

$$L'(a) = \frac{-6(10-a) + 2(a-2)}{16} = \frac{8a-64}{16} = 0 \iff a = 8.$$

General formula: the quantile at level $c_u/(c_u + c_o) = 3/4$. For this uniform, $F(a) = (a-2)/8 = 3/4 \iff a = 8$. ✓ The prediction leans high because under-predicting costs three times more.

(d) Compute the expected pinball loss at $a = 8$ and at the mean $a = 6$. How much does reporting the mean cost?

$L(8) = \frac{3 \cdot 4 + 36}{16} = \frac{48}{16} = 3$; $L(6) = \frac{3 \cdot 16 + 16}{16} = \frac{64}{16} = 4$. Reporting the mean costs $4/3 - 1 \approx 33\%$ more per prediction than the optimal quantile.

1.3 Problem 1.3: Every summary is optimal for its own loss

Problem 1.3 (★★)

For a fixed input x , the output takes two values: $P(Y = 10 \mid x) = 0.3$ and $P(Y = 0 \mid x) = 0.7$.

Tools you need

Expectation and variance of a discrete variable; the squared-loss decomposition [one-read §2.2, mastery box].

(a) Show that for any constant a , $\mathbb{E}[(Y - a)^2 \mid x] = \text{Var}(Y \mid x) + (\mathbb{E}[Y \mid x] - a)^2$.

Write $m = \mathbb{E}[Y \mid x]$ and $Y - a = (Y - m) + (m - a)$. Squaring and taking expectations,

$$\mathbb{E}[(Y - a)^2 \mid x] = \mathbb{E}[(Y - m)^2 \mid x] + 2(m - a) \underbrace{\mathbb{E}[Y - m \mid x]}_{=0} + (m - a)^2 = \text{Var}(Y \mid x) + (m - a)^2. \quad \blacksquare$$

(b) Compute $\mathbb{E}[Y \mid x]$, $\text{Var}(Y \mid x)$, and the expected squared loss at $a = 3$, $a = 5$ and $a = 0$.

$m = 0.3 \times 10 = 3$; $\mathbb{E}[Y^2 \mid x] = 0.3 \times 100 = 30$, so $\text{Var} = 30 - 9 = 21$. By (a): at $a = 3$, 21; at $a = 5$, $21 + 4 = 25$; at $a = 0$, $21 + 9 = 30$.

(c) Find the median, and the expected absolute loss at $a = 0$ and at $a = 3$.

$P(Y \leq 0 \mid x) = 0.7 \geq 0.5$ and $P(Y \geq 0 \mid x) = 1 \geq 0.5$, so the median is 0.

$$\mathbb{E}|Y - 0| = 0.7(0) + 0.3(10) = 3, \quad \mathbb{E}|Y - 3| = 0.7(3) + 0.3(7) = 4.2.$$

(d) Summarise what (b) and (c) show.

Under squared loss the mean (3) beats the median (risk 21 versus 30); under absolute loss the median (0) beats the mean (3 versus 4.2). There is no single “best prediction”: each summary of $P(Y \mid x)$ is optimal for its own loss.

1.4 Problem 1.4: The reject option in general, and what review is worth

Problem 1.4 (★★★)

Binary detection with $p = P(Y = 1 \mid x)$, false-positive cost C_{FP} , false-negative cost C_{FN} , and a review action of fixed cost C_R after which the case is resolved correctly. Correct automatic decisions cost 0.

Tools you need

Conditional risks $R(\text{benign}) = C_{FN}p$, $R(\text{attack}) = C_{FP}(1 - p)$, $R(\text{review}) = C_R$; the lower envelope of the risk lines [one-read §2.2].

(a) Without review, show that the risk of the best action, $\min\{C_{FN}p, C_{FP}(1 - p)\}$, is largest at $p^* = C_{FP}/(C_{FP} + C_{FN})$, with value $C_{FP}C_{FN}/(C_{FP} + C_{FN})$.

$C_{FN}p$ increases with p and $C_{FP}(1-p)$ decreases. Their minimum therefore rises along the first line, then falls along the second, and is largest where they cross: $C_{FN}p = C_{FP}(1-p) \iff p^* = \frac{C_{FP}}{C_{FP}+C_{FN}}$, the familiar threshold. The value there is $C_{FN}p^* = \frac{C_{FP}C_{FN}}{C_{FP}+C_{FN}}$.

(b) Deduce that review is optimal for some p if and only if $C_R < C_{FP}C_{FN}/(C_{FP} + C_{FN})$, and that the regions are then: benign for $p \leq C_R/C_{FN}$, review in between, attack for $p \geq 1 - C_R/C_{FP}$.

Review has constant risk C_R . It beats both automatic actions at some p exactly when the horizontal line C_R passes below the peak of the envelope, i.e. $C_R < C_{FP}C_{FN}/(C_{FP} + C_{FN})$; otherwise it is never optimal. When it is: benign beats review iff $C_{FN}p \leq C_R \iff p \leq C_R/C_{FN}$ (and this is below p^* , so benign also beats attack there); attack beats review iff $C_{FP}(1-p) \leq C_R \iff p \geq 1 - C_R/C_{FP}$ (above p^*). Review wins in between. ■

(c) Apply this to $C_{FP} = 3$, $C_{FN} = 12$, $C_R = 1.5$.

Without review the threshold is $3/15 = 0.2$ and the peak is $36/15 = 2.4 > 1.5$, so review is useful. Regions: benign for $p \leq 1.5/12 = 0.125$; review for $0.125 < p < 0.5$; attack for $p \geq 1 - 1.5/3 = 0.5$.

(d) Suppose the posteriors of incoming flows are uniformly distributed on $[0, 1]$. Compute the expected cost per flow of the optimal policy with and without review.

Integrate the risk of the chosen action over $p \in [0, 1]$ (density 1).

With review:

$$\begin{aligned} & \int_0^{0.125} 12p \, dp + \int_{0.125}^{0.5} 1.5 \, dp + \int_{0.5}^1 3(1-p) \, dp \\ &= 12 \cdot \frac{0.125^2}{2} + 1.5 \times 0.375 + 3 \cdot \frac{0.5^2}{2} = 0.0938 + 0.5625 + 0.375 = 1.031. \end{aligned}$$

Without review:

$$\int_0^{0.2} 12p \, dp + \int_{0.2}^1 3(1-p) \, dp = 12 \cdot \frac{0.2^2}{2} + 3 \cdot \frac{0.8^2}{2} = 0.24 + 0.96 = 1.20.$$

The review option saves 0.169 per flow, about 14%.

1.5 Problem 1.5: Bayes error of two Gaussian classes

Problem 1.5 (★★)

One feature: $X \mid (Y = 0) \sim \mathcal{N}(0, 1)$ and $X \mid (Y = 1) \sim \mathcal{N}(2, 1)$. Values you may need: $\Phi(1) = 0.8413$, $\Phi(1.693) = 0.9548$, $\Phi(0.307) = 0.6205$, and $\Phi(-z) = 1 - \Phi(z)$.

Tools you need

Bayes classifier: compare $\pi_c p(x \mid c)$ [one-read §2.2, 4.3]; the error of a threshold rule is a sum of two normal tail probabilities.

(a) With equal priors, find the Bayes decision rule.

Predict 1 when $\pi_1 \mathcal{N}(x; 2, 1) > \pi_0 \mathcal{N}(x; 0, 1)$. With equal priors, take logs: $-\frac{(x-2)^2}{2} > -\frac{x^2}{2} \iff 4x - 4 > 0 \iff x > 1$. Predict class 1 for $x > 1$: the midpoint of the means.

(b) Compute its error (the Bayes error).

An error happens when a class-0 point exceeds 1 or a class-1 point falls below 1:

$$\frac{1}{2}P(X > 1 \mid 0) + \frac{1}{2}P(X < 1 \mid 1) = \frac{1}{2}(1 - \Phi(1)) + \frac{1}{2}\Phi(-1) = 1 - \Phi(1) = 0.1587.$$

No classifier using this feature can do better.

(c) Now $\pi_0 = 0.8$, $\pi_1 = 0.2$. Find the new boundary.

$\ln 0.2 - \frac{(x-2)^2}{2} = \ln 0.8 - \frac{x^2}{2} \iff \frac{4x-4}{2} = \ln 0.8 - \ln 0.2 = \ln 4 \iff x = 1 + \frac{1}{2} \ln 4 = 1 + \ln 2 = 1.693$.
The boundary moves toward the rare class.

(d) Compute the Bayes error with these priors, and the error if you kept the boundary $x = 1$.

With the boundary 1.693:

$$\begin{aligned} 0.8 P(X > 1.693 | 0) + 0.2 P(X < 1.693 | 1) &= 0.8(1 - 0.9548) + 0.2 \Phi(-0.307) \\ &= 0.8(0.0452) + 0.2(0.3795) = 0.112. \end{aligned}$$

With the boundary 1: $0.8(1 - 0.8413) + 0.2(0.1587) = 0.1587$. Ignoring the priors costs about 0.047, i.e. 42% more errors.

2 Set 2: Learning from data

2.1 Problem 2.1: KNN by hand, and leave-one-out

Problem 2.1 (★)

Six training points, each with a numerical output y (for regression) and a class c (for classification):

x	1	2	4	5	7	8
y	2.0	3.0	5.0	4.0	8.0	9.0
c	0	0	1	1	0	1

The query is $x = 4.6$; distance is $|x - x'|$.

Tools you need

$\hat{f}_k(x) = \frac{1}{k} \sum_{i \in N_k(x)} y^{(i)}$ and $\hat{P}_k(c | x) = \frac{1}{k} \sum_{i \in N_k(x)} \mathbf{1}(c^{(i)} = c)$ [one-read §2.4].

(a) Order the training points by distance to the query, and give the KNN regression prediction for $k = 1, 3, 5, 6$.

Distances: $x = 5$: 0.4; 4: 0.6; 7: 2.4; 2: 2.6; 8: 3.4; 1: 3.6.

k	neighbours	$\hat{f}_k(4.6)$
1	5	4.0
3	5, 4, 7	$(4 + 5 + 8)/3 = 5.67$
5	5, 4, 7, 2, 8	$(4 + 5 + 8 + 3 + 9)/5 = 5.8$
6	all	$31/6 = 5.17$

(b) Give $\hat{P}_k(c = 1 | 4.6)$ and the zero-one decision for the same k .

$k = 1$: 1, class 1. $k = 3$: classes 1, 1, 0, so 2/3, class 1. $k = 5$: classes 1, 1, 0, 0, 1, so 3/5, class 1. $k = 6$: three of each, 1/2: a tie, decided by whatever tie-breaking rule the implementation uses. At $k = N$ the estimate is the overall class frequency, the same at every query.

(c) Compute the leave-one-out (LOO) error of 1-NN classification and the LOO mean squared error of 1-NN regression: predict each point from the other five. Why is the ordinary training error of 1-NN zero?

The nearest *other* point of each x is: $1 \rightarrow 2$, $2 \rightarrow 1$, $4 \rightarrow 5$, $5 \rightarrow 4$, $7 \rightarrow 8$, $8 \rightarrow 7$.

- Classification: the predicted class is wrong only for $x = 7$ (gets class 1 from $x = 8$) and $x = 8$ (gets class 0 from $x = 7$). LOO error $2/6 = 0.33$.
- Regression: each prediction is off by exactly 1 (2 vs 3, 3 vs 2, 5 vs 4, 4 vs 5, 8 vs 9, 9 vs 8), so LOO MSE = 1.

The ordinary training error of 1-NN is 0 because each training point is its own nearest neighbour (distance 0). LOO removes that, which is why it is an honest estimate and the training error is not.

2.2 Problem 2.2: Which distance?

Problem 2.2 (★)

Query $x = (0, 0)$; training points $A = (3, 0)$ and $B = (2, 2)$. The Minkowski distance is $d_p(x, z) = (\sum_j |x_j - z_j|^p)^{1/p}$.

Tools you need

Minkowski family: $p = 1$ Manhattan, $p = 2$ Euclidean [one-read §2.4].

(a) Which point is nearer under $p = 1$, and under $p = 2$?

$p = 1$: $d(A) = 3$, $d(B) = 2 + 2 = 4$: A is nearer. $p = 2$: $d(A) = 3$, $d(B) = \sqrt{8} = 2.83$: B is nearer. The choice of p alone changes the nearest neighbour.

(b) Show that $d_p(x, B) \rightarrow 2 = \max_j |x_j - B_j|$ as $p \rightarrow \infty$, and say which point is nearer in that limit.

$d_p(x, B) = (2^p + 2^p)^{1/p} = (2 \cdot 2^p)^{1/p} = 2 \cdot 2^{1/p} \rightarrow 2$ since $2^{1/p} \rightarrow 1$. In general the largest term dominates the sum, so $d_p \rightarrow \max_j |x_j - z_j|$. In that limit $d(A) = 3$ and $d(B) = 2$: B is nearer. Small p adds up all coordinate differences; large p looks only at the largest one.

2.3 Problem 2.3: Softmax, cross-entropy and their derivatives

Problem 2.3 (★★), parts (c) and (d) looking ahead

A three-class model outputs scores $s = (s_1, s_2, s_3) = (2, 1, -1)$.

Tools you need

$p_c = e^{s_c} / \sum_k e^{s_k}$, $\ell_{CE} = -\ln p_y$, $\sigma(s) = 1/(1 + e^{-s})$ [one-read §2.5, 4.2].

(a) Compute the softmax probabilities, and the cross-entropy loss if the true class is $y = 2$.

$e^2 = 7.389$, $e^1 = 2.718$, $e^{-1} = 0.368$; sum 10.475. Probabilities (0.705, 0.259, 0.035). Loss $-\ln 0.259 = 1.349$.

(b) Show that subtracting 2 from every score leaves the probabilities unchanged, and that for two classes softmax is the sigmoid of the score difference.

$\frac{e^{s_c-2}}{\sum_k e^{s_k-2}} = \frac{e^{-2} e^{s_c}}{e^{-2} \sum_k e^{s_k}} = \frac{e^{s_c}}{\sum_k e^{s_k}}$. With scores (0, -1, -3): 1, 0.368, 0.050, sum 1.418, probabilities (0.705, 0.259, 0.035). ✓ For two classes, dividing top and bottom by e^{s_1} : $p_1 = \frac{e^{s_1}}{e^{s_1} + e^{s_0}} = \frac{1}{1 + e^{-(s_1 - s_0)}} = \sigma(s_1 - s_0)$.

(c) Show that $\partial \ell_{CE} / \partial s_c = p_c - \mathbf{1}(c = y)$, and evaluate this gradient for (a).

Write the loss in terms of scores: $\ell_{CE} = -\ln p_y = -s_y + \ln \sum_k e^{s_k}$. Differentiate with respect to s_c :

$$\frac{\partial \ell_{CE}}{\partial s_c} = -\mathbf{1}(c = y) + \frac{e^{s_c}}{\sum_k e^{s_k}} = p_c - \mathbf{1}(c = y).$$

For $y = 2$: $(0.705, 0.259 - 1, 0.035) = (0.705, -0.741, 0.035)$. Moving against the gradient raises the true class's score and lowers the others in proportion to how much probability they took. The components sum to 0, consistent with (b): shifting all scores together changes nothing.

(d) Binary case with $y \in \{0, 1\}$ and $\hat{p} = \sigma(s)$: show that $\ell_{BCE} = \ln(1 + e^{-s})$ when $y = 1$ and that $\partial \ell_{BCE} / \partial s = \sigma(s) - y$. Evaluate the loss and the derivative at $s = 3$ and $s = -3$ for $y = 1$.

For $y = 1$, $\ell = -\ln \sigma(s) = \ln(1 + e^{-s})$; for $y = 0$, $\ell = -\ln(1 - \sigma(s)) = \ln(1 + e^s)$. Using $\sigma'(s) = \sigma(s)(1 - \sigma(s))$: for $y = 1$, $\frac{d}{ds}[-\ln \sigma] = -(1 - \sigma)$; for $y = 0$, $\frac{d}{ds}[-\ln(1 - \sigma)] = \sigma$. Both are $\sigma(s) - y$.

At $s = 3$: loss $\ln(1 + e^{-3}) = 0.049$, derivative $0.953 - 1 = -0.047$. At $s = -3$: loss $\ln(1 + e^3) = 3.049$, derivative $0.047 - 1 = -0.953$. A confident mistake costs roughly $|s|$ and pushes hard; a confident correct answer costs almost nothing.

2.4 Problem 2.4: Entropy, KL divergence and mutual information

Problem 2.4 (★★)

Work in bits. $p = (0.5, 0.25, 0.25)$ and $q = (0.7, 0.2, 0.1)$. Values you may need: $\log_2 0.7 = -0.5146$, $\log_2 0.2 = -2.3219$, $\log_2 0.1 = -3.3219$, $\log_2 0.3 = -1.7370$, $\log_2 0.8 = -0.3219$.

Tools you need

$H(p) = -\sum p_c \log p_c$, $H(p, q) = -\sum p_c \log q_c$, $D_{KL}(p||q) = \sum p_c \log \frac{p_c}{q_c}$, $H(p, q) = H(p) + D_{KL}(p||q)$, $I(X; Y) = H(Y) - H(Y | X)$ [one-read §2.5].

(a) Compute $H(p)$, $H(p, q)$ and $D_{KL}(p||q)$, and check the identity $H(p, q) = H(p) + D_{KL}(p||q)$.

$H(p) = 0.5(1) + 0.25(2) + 0.25(2) = 1.5$. $H(p, q) = 0.5(0.5146) + 0.25(2.3219) + 0.25(3.3219) = 0.2573 + 0.5805 + 0.8305 = 1.6683$. $D_{KL}(p||q) = 0.5 \log_2 \frac{0.5}{0.7} + 0.25 \log_2 \frac{0.25}{0.2} + 0.25 \log_2 \frac{0.25}{0.1} = -0.2427 + 0.0805 + 0.3305 = 0.1683$. Indeed $1.5 + 0.1683 = 1.6683$. ✓

(b) Compute $D_{KL}(q||p)$. Is the divergence symmetric?

$H(q) = 0.7(0.5146) + 0.2(2.3219) + 0.1(3.3219) = 1.1568$ and $H(q, p) = 0.7(1) + 0.2(2) + 0.1(2) = 1.3$, so $D_{KL}(q||p) = 1.3 - 1.1568 = 0.1432 \neq 0.1683$. Not symmetric: it is not a distance. In cross-entropy training the truth is the *first* argument.

(c) $X, Y \in \{0, 1\}$ with joint probabilities $P(0, 0) = P(1, 1) = 0.4$ and $P(0, 1) = P(1, 0) = 0.1$. Compute $H(Y)$, $H(Y | X)$, $I(X; Y)$, the Bayes error, and the smallest achievable expected cross-entropy.

$P(Y = 1) = 0.5$, so $H(Y) = 1$ bit. Given either value of X , Y agrees with X with probability 0.8, so $H(Y | X) = h(0.8) = 0.8(0.3219) + 0.2(2.3219) = 0.7219$ bits, and $I(X; Y) = 1 - 0.7219 = 0.2781$ bits. The Bayes rule predicts $Y = X$, with error 0.2. The smallest expected cross-entropy is $H(Y | X) = 0.7219$ bits $= 0.7219 \ln 2 = 0.500$ nats: no model can beat it.

(d) A model predicts $\hat{p}(Y = 1 | X = 1) = 0.7$ and $\hat{p}(Y = 1 | X = 0) = 0.3$. Compute its expected cross-entropy with the decomposition $R_{CE} = H(Y | X) + \mathbb{E}_X[D_{KL}]$, then directly.

$D_{KL}(\text{Bern}(0.8)||\text{Bern}(0.7)) = 0.8 \log_2 \frac{0.8}{0.7} + 0.2 \log_2 \frac{0.2}{0.3} = 0.1541 - 0.1170 = 0.0371$; by symmetry the same at $X = 0$. So $R_{CE} = 0.7219 + 0.0371 = 0.7590$ bits.

Directly, at $X = 1$: $0.8(-\log_2 0.7) + 0.2(-\log_2 0.3) = 0.4117 + 0.3474 = 0.7590$; the same at $X = 0$.
 ✓ The model pays 0.037 bits per example for its slightly wrong probabilities.

2.5 Problem 2.5: ERM in closed form

Problem 2.5 (★★)

Empirical risk minimisation for very simple model families, where the minimiser can be written down.

Tools you need

$\hat{\theta} \in \arg \min_{\theta} \hat{R}_{\text{train}}(\theta)$; set the derivative of the training risk to zero [one-read §2.6].

(a) A constant predictor $f(x) = c$ is fitted to $y = (1, 2, 2, 3, 12)$. Show that squared loss gives the sample mean and absolute loss the sample median. Compute both, their training risks, and each one's risk under the other loss.

Squared: $\frac{d}{dc} \frac{1}{N} \sum (y_i - c)^2 = -\frac{2}{N} \sum (y_i - c) = 0 \iff c = \bar{y}$. Absolute: the derivative of $\frac{1}{N} \sum |y_i - c|$ is $\frac{1}{N} (\#\{y_i < c\} - \#\{y_i > c\})$, zero when half the points lie on each side: a median.

Mean = $20/5 = 4$; median = 2.

	squared risk	absolute risk
$c = 4$ (mean)	$(9 + 4 + 4 + 1 + 64)/5 = 16.4$	$(3 + 2 + 2 + 1 + 8)/5 = 3.2$
$c = 2$ (median)	$(1 + 0 + 0 + 1 + 100)/5 = 20.4$	$(1 + 0 + 0 + 1 + 10)/5 = 2.4$

The outlier 12 pulls the mean to 4; the median ignores how far it is.

(b) A constant probability $q = \hat{p}(y = 1)$ is fitted with cross-entropy to N binary labels containing n_1 ones. Show that $\hat{q} = n_1/N$, and compute the minimal training risk for $n_1 = 3$, $N = 10$.

$\hat{R}(q) = -\frac{1}{N} [n_1 \ln q + (N - n_1) \ln(1 - q)]$. Then $\hat{R}'(q) = -\frac{1}{N} [\frac{n_1}{q} - \frac{N - n_1}{1 - q}] = 0 \iff n_1(1 - q) = (N - n_1)q \iff q = n_1/N$. With $\hat{q} = 0.3$: $\hat{R} = -(0.3 \ln 0.3 + 0.7 \ln 0.7) = 0.611$ nats, the entropy of the observed label frequencies.

(c) Fit $f(x) = wx$ (no intercept) by squared loss. Show $\hat{w} = \sum_i x_i y_i / \sum_i x_i^2$ and compute it, with the training MSE, for the data $(1, 2), (2, 3), (3, 7)$.

$\frac{d}{dw} \frac{1}{N} \sum (y_i - wx_i)^2 = -\frac{2}{N} \sum x_i (y_i - wx_i) = 0 \iff \hat{w} = \frac{\sum x_i y_i}{\sum x_i^2}$. Here $\sum x_i y_i = 2 + 6 + 21 = 29$ and $\sum x_i^2 = 14$, so $\hat{w} = 2.071$. Predictions 2.07, 4.14, 6.21; residuals $-0.07, -1.14, 0.79$; MSE = $(0.005 + 1.306 + 0.617)/3 = 0.643$.

(d) Now add an L_2 penalty: minimise $\frac{1}{N} \sum (y_i - wx_i)^2 + \lambda w^2$. Show $\hat{w}_\lambda = \sum x_i y_i / (\sum x_i^2 + N\lambda)$ and compute it for $\lambda = 1$ with its training MSE.

Setting the derivative to zero: $-\frac{2}{N} \sum x_i (y_i - wx_i) + 2\lambda w = 0 \iff \sum x_i y_i = w(\sum x_i^2 + N\lambda)$. With $N = 3$, $\lambda = 1$: $\hat{w}_1 = 29/17 = 1.706$, smaller than 2.071 (shrinkage). Its training MSE is 1.266, higher than 0.643: the penalty deliberately gives up training fit in exchange for a more restricted, lower-variance solution [one-read §3.4].

3 Set 3: Generalisation

3.1 Problem 3.1: Bias and variance of a shrinkage estimator

Problem 3.1 (★★)

Observations $Y_i = \theta + \varepsilon_i$, $i = 1, \dots, N$, with independent noise of mean 0 and variance σ^2 . To predict a new Y , use $\hat{\theta}_\alpha = \alpha \bar{Y}$ with $0 \leq \alpha \leq 1$, where $\bar{Y} = \frac{1}{N} \sum_i Y_i$. The case $\alpha = 1$ is the sample mean; $\alpha < 1$ shrinks it toward 0, like a penalty.

Tools you need

Bias, variance and $\text{MSE} = \text{bias}^2 + \text{variance}$ [one-read §3.3]; $\mathbb{E}\bar{Y} = \theta$, $\text{Var}\bar{Y} = \sigma^2/N$.

(a) Compute the bias and the variance of $\hat{\theta}_\alpha$.

$\mathbb{E}\hat{\theta}_\alpha = \alpha\theta$, so bias $= \alpha\theta - \theta = -(1 - \alpha)\theta$. $\text{Var}\hat{\theta}_\alpha = \alpha^2 \text{Var}\bar{Y} = \alpha^2\sigma^2/N$. Shrinking introduces bias and removes variance.

(b) Show that $\text{MSE}(\alpha) = \mathbb{E}[(\hat{\theta}_\alpha - \theta)^2] = (1 - \alpha)^2\theta^2 + \alpha^2\sigma^2/N$, and find the α^* that minimises it.

Add and subtract the mean $\alpha\theta$: $\mathbb{E}[(\hat{\theta}_\alpha - \theta)^2] = \mathbb{E}[(\hat{\theta}_\alpha - \alpha\theta)^2] + (\alpha\theta - \theta)^2$ (the cross term has expectation 0), i.e. variance plus squared bias, which is the stated formula. Then

$$\frac{d\text{MSE}}{d\alpha} = -2(1 - \alpha)\theta^2 + 2\alpha\frac{\sigma^2}{N} = 0 \iff \alpha^* = \frac{\theta^2}{\theta^2 + \sigma^2/N}.$$

$\alpha^* < 1$ whenever there is noise: a little bias is worth it. As $N \rightarrow \infty$, $\alpha^* \rightarrow 1$: with enough data, no shrinkage is needed.

(c) With $\theta = 2$, $\sigma^2 = 4$, $N = 4$: compute α^* , $\text{MSE}(1)$, $\text{MSE}(\alpha^*)$ and $\text{MSE}(0.5)$.

$\sigma^2/N = 1$, so $\alpha^* = 4/5 = 0.8$. $\text{MSE}(1) = 0 + 1 = 1$; $\text{MSE}(0.8) = 0.04 \times 4 + 0.64 \times 1 = 0.8$; $\text{MSE}(0.5) = 0.25 \times 4 + 0.25 \times 1 = 1.25$. Too little shrinkage leaves variance, too much adds bias: the U-shape in one dimension.

(d) Show that the expected squared error in predicting a new, independent $Y = \theta + \varepsilon$ is $\sigma^2 + \text{MSE}(\alpha)$, and evaluate it at $\alpha = 1$ and α^* .

$Y - \hat{\theta}_\alpha = \varepsilon + (\theta - \hat{\theta}_\alpha)$, and the new noise ε is independent of the data, so the cross term vanishes: $\mathbb{E}[(Y - \hat{\theta}_\alpha)^2] = \sigma^2 + \text{MSE}(\alpha)$. This is the decomposition noise + bias² + variance. At $\alpha = 1$: 5; at α^* : 4.8. The $\sigma^2 = 4$ part is irreducible.

(e) Why can α^* not be computed in practice, and how would it be chosen?

α^* depends on θ , the unknown quantity being estimated. In practice α (like a regularisation strength λ) is a hyperparameter, chosen by comparing candidate values on validation data or by cross-validation [one-read §2.7].

3.2 Problem 3.2: Bias and variance of KNN regression

Problem 3.2 (★★)

True function $f^*(x) = x^2$; observations $y_i = f^*(x_i) + \varepsilon_i$ with independent noise of variance $\sigma^2 = 0.01$, at fixed inputs $x = \pm 0.1, \pm 0.2, \pm 0.3, \pm 0.4, \pm 0.5$ (one observation each). Query: $x = 0$, where $f^*(0) = 0$. Only the noise is random (the inputs are fixed).

Tools you need

$\hat{f}_k(0) = \frac{1}{k} \sum_{i \in N_k(0)} y_i$; bias = $\mathbb{E}\hat{f}_k(0) - f^*(0)$; variance of an average of k independent noises is σ^2/k [one-read §3.3].

(a) For even k , show that $\text{Bias}(0) = \frac{1}{k} \sum_{i \in N_k(0)} x_i^2$ and $\text{Var}(0) = \sigma^2/k$.

The k nearest inputs are $\pm 0.1, \dots, \pm 0.1(k/2)$. $\mathbb{E}\hat{f}_k(0) = \frac{1}{k} \sum_i \mathbb{E}y_i = \frac{1}{k} \sum_i x_i^2$; subtracting $f^*(0) = 0$ gives the bias. The prediction is an average of k independent noise terms, each of variance σ^2 , so its variance is $k\sigma^2/k^2 = \sigma^2/k$.

(b) Compute bias, variance and $\text{MSE} = \text{bias}^2 + \text{variance}$ for $k = 2, 4, 6, 8, 10$, and find the best k .

k	neighbours	bias	bias ²	variance $0.01/k$	MSE
2	± 0.1	0.010	0.000100	0.00500	0.00510
4	$\pm 0.1, \pm 0.2$	0.025	0.000625	0.00250	0.00313
6	$\pm 0.1, \pm 0.3$	0.0467	0.00218	0.00167	0.00384
8	$\pm 0.1, \pm 0.4$	0.075	0.00563	0.00125	0.00688
10	all	0.110	0.0121	0.00100	0.0131

(Example: for $k = 6$, bias = $2(0.01 + 0.04 + 0.09)/6 = 0.0467$.) Best: $k = 4$. Small k is noisy, large k averages in points where x^2 is far from 0: the U-shape with $1/k$ playing the role of capacity.

(c) What is the expected squared prediction error for a new observation at $x = 0$ with the best k ?

$\sigma^2 + \text{MSE} = 0.01 + 0.00313 = 0.0131$. Even a perfect predictor would pay 0.01.

3.3 Problem 3.3: How precise is a test score?

Problem 3.3 (★★)

A *fixed* classifier has true error rate R (zero-one loss). It is evaluated on N independent test examples. Chebyshev's inequality: for any variable Z with mean μ and variance v , $P(|Z - \mu| \geq t) \leq v/t^2$.

Tools you need

For a fixed model, $\mathbb{E}\hat{R} = R$ and $\text{Var}\hat{R} = \sigma_\ell^2/N$ [one-read §2.6, mastery box].

(a) Show that $\hat{R}_{\text{test}}$ is unbiased with variance $R(1 - R)/N$.

Each test loss $\ell_i = \mathbf{1}(\text{error on } i)$ is Bernoulli with mean R and variance $R(1 - R)$. The average of N independent such terms has mean R and variance $R(1 - R)/N$.

(b) For $R = 0.08$ and $N = 2000$, compute the standard error and an approximate 95% interval (two standard errors).

$\text{SE} = \sqrt{0.08 \times 0.92/2000} = \sqrt{0.0000368} = 0.0061$. Interval: 0.08 ± 0.012 , i.e. about $[0.068, 0.092]$.

(c) How many test examples make the standard error at most 0.002?

$R(1 - R)/N \leq 0.002^2 \iff N \geq 0.0736/0.000004 = 18,400$. Tripling the precision costs nine times the data.

(d) A recall of 0.9 is measured on a test set containing 50 positives. What is its standard error?

Recall is an average over the positives only, so $N = 50$: $SE = \sqrt{0.9 \times 0.1/50} = 0.042$, i.e. roughly 0.9 ± 0.085 . Rare-class metrics are noisy.

(e) Use Chebyshev to bound $P(|\hat{R}_{\text{test}} - R| \geq 0.02)$ in the setting of (b).

$P \leq \frac{0.0000368}{0.02^2} = 0.092$. A guarantee that holds for any distribution, so it is loose (the normal approximation gives about 0.001), but it needs no assumption.

3.4 Problem 3.4: Selection noise

Problem 3.4 (★★)

Validation errors are $\hat{R}_j = R_j + \varepsilon_j$ with independent $\varepsilon_j \sim \mathcal{N}(0, \sigma^2)$. You may use: $\mathbb{E} \min(Z_1, Z_2) = -1/\sqrt{\pi}$ and $\mathbb{E} \min(Z_1, Z_2, Z_3) = -3/(2\sqrt{\pi})$ for independent standard normals; $\Phi(0.354) = 0.638$, $\Phi(1.645) = 0.95$.

Tools you need

Winner's curse $\mathbb{E} \min_j \hat{R}_j \leq \min_j R_j$ [one-read §3.1, 3.5]; the difference of two independent normals.

(a) Three configurations all have $R = 0.20$ and $\sigma = 0.02$. What is the expected validation error reported for the selected one?

$\mathbb{E} \min_j \hat{R}_j = R + \sigma \mathbb{E} \min(Z_1, Z_2, Z_3) = 0.20 - 0.02 \times \frac{3}{2\sqrt{\pi}} = 0.20 - 0.0169 = 0.183$. With two configurations it was $0.20 - 0.0113$: more candidates, more optimism.

(b) Two configurations have true errors 0.20 and 0.21, with $\sigma = 0.02$ each. What is the probability of selecting the worse one?

The worse one is chosen when $\hat{R}_2 < \hat{R}_1$, i.e. $\varepsilon_2 - \varepsilon_1 < -0.01$. The difference is $\mathcal{N}(0, 2\sigma^2)$ with standard deviation $0.02\sqrt{2} = 0.0283$, so the probability is $\Phi(-0.01/0.0283) = \Phi(-0.354) = 1 - 0.638 = 0.362$. More than a third of the time.

(c) Take $\sigma^2 \approx 0.16/N$ for both, where N is the validation-set size ($\sigma = 0.02$ is $N = 400$). How large must N be for the probability in (b) to fall below 5%?

We need $0.01/(\sigma\sqrt{2}) \geq 1.645$, i.e. $\sigma \leq 0.01/(1.645\sqrt{2}) = 0.00430$, so $0.16/N \leq 1.848 \times 10^{-5}$ and $N \geq 8,660$. To tell apart models one point of error apart, you need validation sets in the thousands.

3.5 Problem 3.5: L_1 and L_2 in numbers

Problem 3.5 (★★)

One-dimensional penalised problems with a least-squares estimate z :

$$\text{lasso: } \min_{\theta} \frac{1}{2}(\theta - z)^2 + \lambda|\theta|, \quad \text{ridge: } \min_{\theta} \frac{1}{2}(\theta - z)^2 + \lambda\theta^2.$$

Tools you need

Regularised ERM, L_1 versus L_2 [one-read §3.4]; for the lasso, treat $\theta > 0$, $\theta < 0$ and $\theta = 0$ separately.

(a) Show that the lasso solution is $\text{sign}(z) \max(|z| - \lambda, 0)$ and the ridge solution is $z/(1 + 2\lambda)$. Compute both for $z = 0.8$ with $\lambda = 0.5$ and $\lambda = 1$.

Lasso. For $\theta > 0$ the derivative is $\theta - z + \lambda$, zero at $\theta = z - \lambda$, valid only if $z > \lambda$; for $\theta < 0$, $\theta = z + \lambda$, valid if $z < -\lambda$; if $|z| \leq \lambda$ neither works and the minimum is at the kink, $\theta = 0$. Together: $\text{sign}(z) \max(|z| - \lambda, 0)$, “soft thresholding”. *Ridge.* $\theta - z + 2\lambda\theta = 0 \iff \theta = z/(1 + 2\lambda)$.

	$\lambda = 0.5$	$\lambda = 1$
lasso	0.3	0
ridge	0.4	0.267

(b) With an orthonormal design each coefficient decouples. For least-squares coefficients $z = (3, -0.5, 1.2, 0.1)$ and $\lambda = 0.6$, compute both solutions and count the non-zero coefficients.

Lasso: subtract 0.6 from each absolute value, floor at 0: $(2.4, 0, 0.6, 0)$: **two** non-zeros. *Ridge:* divide by $1 + 1.2 = 2.2$: $(1.364, -0.227, 0.545, 0.045)$: all **four** non-zero. The lasso removes the two small coefficients entirely; ridge shrinks everything proportionally.

(c) Constrained form: minimise $(\theta_1 - 2)^2 + (\theta_2 - 1)^2$ subject to $|\theta_1| + |\theta_2| \leq 1$, then subject to $\theta_1^2 + \theta_2^2 \leq 1$.

The unconstrained optimum $(2, 1)$ lies outside both regions, so the solution is on the boundary.

L_1 : the nearest point of the diamond to $(2, 1)$ has $\theta_1, \theta_2 \geq 0$ and $\theta_1 + \theta_2 = 1$. Substitute $\theta_2 = 1 - \theta_1$: $g(\theta_1) = (\theta_1 - 2)^2 + \theta_1^2$, $g' = 4\theta_1 - 4 = 0 \Rightarrow \theta_1 = 1$, $\theta_2 = 0$. The solution $(1, 0)$ is a *corner*: sparse.

L_2 : the nearest point of the disc is the radial projection $(2, 1)/\sqrt{5} = (0.894, 0.447)$: both non-zero. Exactly the picture in [one-read §3.4].

3.6 Problem 3.6: Early stopping and patience

Problem 3.6 (★)

Validation loss after each epoch of training (at most 12 epochs):

epoch	1	2	3	4	5	6	7	8	9	10	11	12
loss	.90	.70	.58	.52	.50	.51	.53	.52	.49	.47	.46	.48

Rule: stop after p consecutive epochs without improving on the best loss so far; return the best checkpoint.

Tools you need

$t^* = \arg \min_t \hat{R}_{\text{val}}(\hat{\theta}_t)$ with patience [one-read §3.4].

(a) For patience $p = 2, 3, 4$: when does training stop, which checkpoint is returned, and how many epochs were run?

The best loss is 0.50 at epoch 5; epochs 6, 7, 8 do not improve on it; epoch 9 does (0.49).

p	stops after epoch	returns	epochs run
2	7	epoch 5, loss 0.50	7
3	8	epoch 5, loss 0.50	8
4	12 (limit reached)	epoch 11, loss 0.46	12

(b) Interpret the result.

The loss rises after epoch 5, then descends again: an epoch-wise second descent [one-read §3.3]. A short patience stops in the bump and returns 0.50; patience 4 survives the bump and finds 0.46, at the price of 12 epochs instead of 7.

3.7 Problem 3.7: Leakage through a scaler

Problem 3.7 (★)

A one-feature anomaly detector standardises its input and flags $|z| > 3$. Training values: 2, 4, 6, 8. A test value: 20. Standard deviations use the $1/N$ formula.

Tools you need

$\hat{\mu}_j, \hat{\sigma}_j$ fitted on training data only [one-read §3.6].

(a) Fit the scaler on the training data, transform the test value, and decide.

$\hat{\mu} = 5, \hat{\sigma}^2 = (9 + 1 + 1 + 9)/4 = 5, \hat{\sigma} = 2.236. z = (20 - 5)/2.236 = 6.71 > 3$: **flagged**.

(b) Now fit the scaler on all five values (the leaky way). Transform the test value and decide.

$\hat{\mu} = 40/5 = 8, \hat{\sigma}^2 = (36 + 16 + 4 + 0 + 144)/5 = 40, \hat{\sigma} = 6.32. z = (20 - 8)/6.32 = 1.90 < 3$: **not flagged**. The test point inflated the scale used to judge it, so the anomaly hid itself. Leakage does not only flatter scores; it can change decisions.

3.8 Problem 3.8: Precision depends on the base rate

Problem 3.8 (★★)

A detector has recall (true-positive rate) 0.9 and false-positive rate 0.02, whatever the prevalence π of attacks.

Tools you need

Recall $TP/(TP+FN)$, precision $TP/(TP+FP)$, false-positive rate $FP/(FP+TN)$, F_1 [one-read §3.6]; Bayes' rule.

(a) Show that precision =
$$\frac{\text{Rec} \cdot \pi}{\text{Rec} \cdot \pi + \text{FPR} \cdot (1 - \pi)}.$$

Among n flows, $n\pi$ are attacks and $n(1 - \pi)$ benign. Then $TP = \text{Rec} \cdot n\pi$ and $FP = \text{FPR} \cdot n(1 - \pi)$; divide TP by $TP + FP$ and cancel n . (It is Bayes' rule for $P(\text{attack} | \text{alarm})$.)

(b) Compute precision, F_1 and the number of alarms per 100,000 flows for $\pi = 0.1, 0.01, 0.001$.

π	precision	F_1	alarms per 100,000
0.1	$0.09/0.108 = 0.833$	0.865	10,800
0.01	$0.009/0.0288 = 0.313$	0.464	2,880
0.001	$0.0009/0.02088 = 0.043$	0.082	2,088

The same detector goes from 83% precision to 4%: at $\pi = 0.001$, 96 of every 100 alarms are false.

(c) What false-positive rate is needed for a precision of 0.5 at $\pi = 0.001$?

Precision 0.5 means $TP = FP$: $0.9 \times 0.001 = \text{FPR} \times 0.999$, so $\text{FPR} = 0.0009$, about 22 times smaller than 0.02.

4 Set 4: Discriminative and generative classification

4.1 Problem 4.1: Logistic regression by hand

Problem 4.1 (★★), part (e) looking ahead

A logistic regression model with $w = (1.5, -2)$ and $b = 0.5$: $s(x) = w^\top x + b$, $\hat{p}(y = 1 \mid x) = \sigma(s(x))$.

Tools you need

Sigmoid, cross-entropy, log-odds, probability thresholds as score thresholds [one-read §4.2]; $\partial \ell_{BCE} / \partial s = \sigma(s) - y$ (Problem 2.3).

(a) Compute the score and the probability for $x_A = (1, 1)$ and $x_B = (2, 0.5)$.

$s(x_A) = 1.5 - 2 + 0.5 = 0$, so $\hat{p} = \sigma(0) = 0.5$. $s(x_B) = 3 - 1 + 0.5 = 2.5$, so $\hat{p} = 1/(1 + e^{-2.5}) = 0.924$.

(b) The labels are $y_A = 1$ and $y_B = 0$. Compute both cross-entropy losses.

x_A : $-\ln 0.5 = 0.693$. x_B : the model gives class 0 probability 0.076, so the loss is $-\ln 0.076 = 2.58$, a confident mistake.

(c) Show that the log-odds $\ln \frac{\hat{p}}{1-\hat{p}}$ equal the score, and deduce what a one-unit increase in x_1 does to the odds.

$\hat{p} = \frac{1}{1+e^{-s}}$ and $1 - \hat{p} = \frac{e^{-s}}{1+e^{-s}}$, so $\frac{\hat{p}}{1-\hat{p}} = e^s$ and $\ln \frac{\hat{p}}{1-\hat{p}} = s$. Increasing x_1 by 1 adds $w_1 = 1.5$ to the log-odds, i.e. multiplies the odds by $e^{1.5} = 4.48$, whatever the other feature.

(d) Write the decision boundary for a cost threshold of 0.8 on $\hat{p}$.

$\hat{p} > 0.8 \iff s > \ln \frac{0.8}{0.2} = \ln 4 = 1.386 \iff 1.5x_1 - 2x_2 + 0.5 > 1.386 \iff 1.5x_1 - 2x_2 > 0.886$. A line parallel to the 0.5 boundary $1.5x_1 - 2x_2 = -0.5$, shifted toward the positive side.

(e) Take one gradient step on x_B alone with learning rate 0.1: use $\nabla_w \ell = (\sigma(s) - y)x$ and $\partial \ell / \partial b = \sigma(s) - y$. What is the new probability for x_B ?

$\sigma(s) - y = 0.924 - 0 = 0.924$. Gradient: $0.924 \times (2, 0.5) = (1.848, 0.462)$ for w and 0.924 for b . Update: $w \leftarrow (1.5 - 0.185, -2 - 0.046) = (1.315, -2.046)$, $b \leftarrow 0.5 - 0.092 = 0.408$. New score $2.630 - 1.023 + 0.408 = 2.015$, probability 0.882: down from 0.924, toward the correct label 0.

4.2 Problem 4.2: Maximum likelihood for Gaussian classes

Problem 4.2 (★★★)

One class with observations $x_1, \dots, x_N$ modelled as $\mathcal{N}(\mu, \sigma^2)$, and C classes with counts $N_1, \dots, N_C$ ($\sum_c N_c = N$) for the priors.

Tools you need

Log-likelihood, setting derivatives to zero, a Lagrange multiplier for $\sum_c \pi_c = 1$ [one-read §4.3, 4.4].

(a) Derive $\hat{\mu} = \frac{1}{N} \sum_i x_i$ and $\hat{\sigma}^2 = \frac{1}{N} \sum_i (x_i - \hat{\mu})^2$.

The log-likelihood is $\ell(\mu, \sigma^2) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2$.

$$\text{in } \mu \quad \frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_i (x_i - \mu) = 0 \iff \hat{\mu} = \frac{1}{N} \sum_i x_i$$

$$\text{in } \sigma^2 \quad \frac{\partial \ell}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (x_i - \hat{\mu})^2 = 0 \iff \hat{\sigma}^2 = \frac{1}{N} \sum_i (x_i - \hat{\mu})^2$$

(b) Show that maximising $\sum_c N_c \ln \pi_c$ subject to $\sum_c \pi_c = 1$ gives $\hat{\pi}_c = N_c/N$.

Maximise $\mathcal{L} = \sum_c N_c \ln \pi_c - \lambda(\sum_c \pi_c - 1)$. Then $\partial \mathcal{L} / \partial \pi_c = N_c / \pi_c - \lambda = 0$, so $\pi_c = N_c / \lambda$; the constraint gives $\sum_c N_c / \lambda = 1$, so $\lambda = N$ and $\hat{\pi}_c = N_c / N$.

(c) Show that $\mathbb{E}[\hat{\sigma}^2] = \frac{N-1}{N} \sigma^2$ when the x_i are independent $\mathcal{N}(\mu, \sigma^2)$. Hint: first show $\sum_i (x_i - \bar{x})^2 = \sum_i (x_i - \mu)^2 - N(\bar{x} - \mu)^2$.

Write $x_i - \bar{x} = (x_i - \mu) - (\bar{x} - \mu)$ and expand:

$$\sum_i (x_i - \bar{x})^2 = \sum_i (x_i - \mu)^2 - 2(\bar{x} - \mu) \underbrace{\sum_i (x_i - \mu)}_{=N(\bar{x} - \mu)} + N(\bar{x} - \mu)^2 = \sum_i (x_i - \mu)^2 - N(\bar{x} - \mu)^2.$$

Take expectations: $\mathbb{E} \sum_i (x_i - \mu)^2 = N\sigma^2$ and $\mathbb{E}(\bar{x} - \mu)^2 = \text{Var } \bar{x} = \sigma^2 / N$, so $\mathbb{E} \sum_i (x_i - \bar{x})^2 = (N-1)\sigma^2$ and $\mathbb{E} \hat{\sigma}^2 = \frac{N-1}{N} \sigma^2$. The ML estimate is slightly too small, because $\bar{x}$ was fitted to the same points. ■

(d) For a class with observations 2, 4, 9, compute $\hat{\mu}$, the ML variance, and the unbiased variance.

$\hat{\mu} = 5$; deviations $-3, -1, 4$; $\sum = 26$. ML: $26/3 = 8.67$. Unbiased: $26/2 = 13$. With 3 points the difference is large; with hundreds it is negligible.

4.3 Problem 4.3: LDA and QDA disagree

Problem 4.3 (☆☆)

Two classes in $\mathbb{R}^2$, four points each:

class 0: (0, 0), (2, 0), (0, 2), (2, 2); class 1: (4, 4), (6, 4), (5, 5), (5, 3).

Use ML estimates. The pooled covariance for LDA is $\hat{\Sigma} = \sum_c \frac{N_c}{N} \hat{\Sigma}_c$. Query point: $x = (3, 3)$.

Tools you need

Gaussian ML estimates, QDA score $-\frac{1}{2}(x - \hat{\mu}_c)^\top \hat{\Sigma}_c^{-1}(x - \hat{\mu}_c) - \frac{1}{2} \ln |\hat{\Sigma}_c| + \ln \hat{\pi}_c$, LDA score $x^\top \hat{\Sigma}^{-1} \hat{\mu}_c - \frac{1}{2} \hat{\mu}_c^\top \hat{\Sigma}^{-1} \hat{\mu}_c + \ln \hat{\pi}_c$ [one-read §4.4].

(a) Compute $\hat{\pi}_c$, $\hat{\mu}_c$, $\hat{\Sigma}_c$ and the pooled $\hat{\Sigma}$.

$\hat{\pi}_0 = \hat{\pi}_1 = 0.5$. $\hat{\mu}_0 = (1, 1)$, $\hat{\mu}_1 = (5, 4)$.

Class 0 deviations $(\pm 1, \pm 1)$: each coordinate has variance $(1 + 1 + 1 + 1)/4 = 1$; the cross products $(+1, -1, -1, +1)$ average to 0. So $\hat{\Sigma}_0 = I$.

Class 1 deviations $(-1, 0), (1, 0), (0, 1), (0, -1)$: variances $2/4 = 0.5$ each, cross products all 0. So $\hat{\Sigma}_1 = 0.5 I$.

Pooled: $\hat{\Sigma} = 0.5 I + 0.5 \times 0.5 I = 0.75 I$.

(b) QDA: compute both scores at (3, 3), the prediction and $\hat{p}(y = 1 | x)$.

Class 0: $x - \hat{\mu}_0 = (2, 2)$, $\hat{\Sigma}_0^{-1} = I$, quadratic term 8, $\ln |I| = 0$: $s_0 = -4 + \ln 0.5 = -4.693$. Class 1: $x - \hat{\mu}_1 = (-2, -1)$, $\hat{\Sigma}_1^{-1} = 2I$, quadratic term $2 \times 5 = 10$; $|\hat{\Sigma}_1| = 0.25$, so $-\frac{1}{2} \ln 0.25 = +0.693$: $s_1 = -5 + 0.693 - 0.693 = -5.000$. $s_0 > s_1$: **class 0**, with $\hat{p}(y = 1 | x) = 1/(1 + e^{s_0 - s_1}) = 1/(1 + e^{0.307}) = 0.424$.

(c) LDA: compute both scores at (3, 3) (the common $\ln 0.5$ may be dropped), the prediction and $\hat{p}(y = 1 | x)$.

$\hat{\Sigma}^{-1} = \frac{4}{3} I$. $s_0 = \frac{4}{3}(3+3) - \frac{1}{2} \cdot \frac{4}{3} \cdot 2 = 8 - 1.333 = 6.667$. $s_1 = \frac{4}{3}(15+12) - \frac{1}{2} \cdot \frac{4}{3} \cdot 41 = 36 - 27.333 = 8.667$. $s_1 > s_0$: **class 1**, with $\hat{p}(y = 1 | x) = 1/(1 + e^{-2}) = 0.881$. The two models disagree at the same point.

(d) Find both decision boundaries and explain the disagreement.

LDA: $s_1 = s_0 \iff \frac{4}{3} x^\top (\hat{\mu}_1 - \hat{\mu}_0) = \frac{1}{2} \cdot \frac{4}{3} (\|\hat{\mu}_1\|^2 - \|\hat{\mu}_0\|^2) \iff 4x_1 + 3x_2 = 19.5$. At (3, 3): $21 > 19.5$, the class 1 side.

QDA: with $s_0 = -\frac{1}{2}\|x - \hat{\mu}_0\|^2 + \ln 0.5$ and $s_1 = -\|x - \hat{\mu}_1\|^2 + \ln 2 + \ln 0.5$, the condition $s_1 > s_0$ is $\|x - \hat{\mu}_1\|^2 - \frac{1}{2}\|x - \hat{\mu}_0\|^2 < \ln 2$. Expanding and multiplying by 2:

$$x_1^2 - 18x_1 + x_2^2 - 14x_2 + 80 < 2 \ln 2 \iff (x_1 - 9)^2 + (x_2 - 7)^2 < 51.39,$$

the inside of a circle centred at (9, 7) with radius 7.17. The point (3, 3) has $(3 - 9)^2 + (3 - 7)^2 = 52 > 51.39$: just outside, so class 0.

Why. QDA knows class 1 is tight (variance 0.5) and class 0 spread out (variance 1). Measured in each class's own units, (3, 3) is $\sqrt{10} = 3.2$ standard units from $\hat{\mu}_1$ but only $\sqrt{8} = 2.8$ from $\hat{\mu}_0$. LDA pools the two into one variance 0.75, loses that information, and just compares Euclidean-type distances, where (3, 3) is closer to $\hat{\mu}_1$ ($\sqrt{5}$ versus $\sqrt{8}$).

(e) Count the parameters estimated by LDA and by QDA for $d = 2$, $C = 2$, and for $d = 50$, $C = 3$.

A covariance matrix has $d(d + 1)/2$ free parameters; priors $C - 1$; means Cd .

	LDA	QDA
$d = 2, C = 2$	$4 + 3 + 1 = 8$	$4 + 2 \times 3 + 1 = 11$
$d = 50, C = 3$	$150 + 1275 + 2 = 1,427$	$150 + 3 \times 1275 + 2 = 3,977$

4.4 Problem 4.4: Gaussian Naive Bayes, and why variances need smoothing

Problem 4.4 (★★)

Two features (log-duration x_1 , log-bytes x_2), independent given the class:

	$x_1 y$	$x_2 y$	prior
benign	$\mathcal{N}(3, 1)$	$\mathcal{N}(6, 4)$	0.9
attack	$\mathcal{N}(1, 0.25)$	$\mathcal{N}(4, 1)$	0.1

Use $\ln \mathcal{N}(x; \mu, \sigma^2) = -\frac{1}{2} \ln(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}$ and $\ln(2\pi) = 1.8379$.

Tools you need

Naive Bayes score $\ln \hat{\pi}_c + \sum_j \ln p(x_j | c)$; variance smoothing $\hat{\sigma}^2 = \hat{\sigma}^2 + \lambda s^2$ [one-read §4.5].

(a) For $x = (1.5, 5)$, compute every log-density, both class scores, and $\hat{p}(\text{attack} | x)$.

	$\ln p(x_1 c)$	$\ln p(x_2 c)$	$\ln \pi_c$
benign	$-0.919 - 1.125 = -2.044$	$-1.612 - 0.125 = -1.737$	-0.105
attack	$-0.226 - 0.5 = -0.726$	$-0.919 - 0.5 = -1.419$	-2.303

Scores: benign -3.886 , attack -4.447 . $\hat{p}(\text{attack} | x) = 1/(1 + e^{-4.447+3.886}) = 1/(1 + e^{0.561}) = 0.363$. Both features favour attack, but not enough to overcome a 9:1 prior.

(b) Suppose the fitted attack variance of x_1 were 0.001 (a nearly constant feature in a tiny sample). Compute $\ln p(x_1 = 1.2 | \text{attack})$. Then smooth it with $\lambda = 0.1$ and an overall variance $s^2 = 2$, and recompute.

Unsmoothed: $-\frac{1}{2} \ln(2\pi \times 0.001) - \frac{0.04}{0.002} = 2.53 - 20 = -17.47$. A value only 0.2 from the mean is treated as almost impossible, and this single term overwhelms every other feature.

Smoothed: $\hat{\sigma}^2 = 0.001 + 0.1 \times 2 = 0.201$, so $-\frac{1}{2} \ln(2\pi \times 0.201) - \frac{0.04}{0.402} = -0.117 - 0.100 = -0.216$. The feature now gives reasonable, bounded evidence.

4.5 Problem 4.5: Multinomial Naive Bayes for spam

Problem 4.5 (★★)

Vocabulary (login, password, verify, invoice, meeting). Training token counts: spam (8, 6, 5, 1, 0), total 20; ham (1, 1, 0, 4, 9), total 15. Priors: spam 0.4, ham 0.6. New message: “verify password password meeting”, count vector $x = (0, 2, 1, 0, 1)$.

Tools you need

$$p(x | c) \propto \prod_j \theta_{cj}^{x_j}, \hat{\theta}_{cj} = \frac{n_{cj} + \alpha}{\sum_\ell n_{c\ell} + \alpha V} \text{ [one-read §4.5].}$$

(a) With $\alpha = 1$, compute the smoothed token probabilities of each class.

$V = 5$, so each denominator adds $\alpha V = 5$.

$$\text{spam } \hat{\theta} = (9, 7, 6, 2, 1)/25 = (0.36, 0.28, 0.24, 0.08, 0.04)$$

$$\text{ham } \hat{\theta} = (2, 2, 1, 5, 10)/20 = (0.10, 0.10, 0.05, 0.25, 0.50)$$

(b) Compute both log-scores and $\hat{p}(\text{spam} | x)$.

Spam: $\ln 0.4 + 2 \ln 0.28 + \ln 0.24 + \ln 0.04 = -0.916 - 2.546 - 1.427 - 3.219 = -8.108$. Ham: $\ln 0.6 + 2 \ln 0.10 + \ln 0.05 + \ln 0.50 = -0.511 - 4.605 - 2.996 - 0.693 = -8.805$. $\hat{p}(\text{spam} | x) = 1/(1 + e^{-8.805+8.108}) = 1/(1 + e^{-0.697}) = 0.667$. “password” twice and “verify” outweigh “meeting”.

(c) What happens without smoothing?

Spam has $n = 0$ for “meeting”, so $\theta_{\text{spam}, \text{meeting}} = 0$ and the spam likelihood is 0; ham has $n = 0$ for “verify”, so the ham likelihood is 0 too. The posterior is 0/0: undefined. One unseen word per class destroys the model.

(d) What does the posterior tend to as $\alpha \rightarrow \infty$?

$\hat{\theta}_{cj} \rightarrow \alpha/(\alpha V) = 1/V$ for every word and class, so both class likelihoods become equal and the posterior tends to the prior, 0.4. Smoothing interpolates between the data ($\alpha \rightarrow 0$) and ignoring the message ($\alpha \rightarrow \infty$).

4.6 Problem 4.6: Prior shift in odds form

Problem 4.6 (★★)

A binary classifier was trained on data resampled to balance the classes ($\pi_t = 0.5$). In deployment, attacks are $\pi_d = 0.02$ of flows. Assume prior shift and a calibrated training posterior $r_t(x)$.

Tools you need

Posterior odds = prior odds $\times$ likelihood ratio; the prior-shift correction [one-read §4.7].

(a) Derive $\frac{r_d}{1-r_d} = \frac{\pi_d/(1-\pi_d)}{\pi_t/(1-\pi_t)} \cdot \frac{r_t}{1-r_t}$.

By Bayes' rule, $\frac{r}{1-r} = \frac{p(y=1|x)}{p(y=0|x)} = \frac{\pi}{1-\pi} \cdot \frac{p(x|1)}{p(x|0)}$. Under prior shift the likelihood ratio $\Lambda(x) = p(x|1)/p(x|0)$ is the same in training and deployment. Write the identity for both and divide: Λ cancels, leaving the stated formula.

(b) Compute r_d for $r_t = 0.5$ and $r_t = 0.9$.

The prior-odds ratio is $\frac{0.02/0.98}{0.5/0.5} = 0.0204$. $r_t = 0.5$: odds $1 \rightarrow 0.0204$, so $r_d = 0.020$. $r_t = 0.9$: odds $9 \rightarrow 0.184$, so $r_d = 0.155$. A “90% sure” output on balanced training data means about 16% in the real stream.

(c) In deployment, alerts are raised when $r_d > 0.2$. What threshold on r_t is equivalent?

$r_d > 0.2 \iff \text{deployment odds} > 0.25 \iff \text{training odds} > 0.25/0.0204 = 12.25 \iff r_t > 12.25/13.25 = 0.925$.

(d) A generative model gives likelihood ratio $\Lambda(x) = 6$ for some flow. Compute its posterior under $\pi = 0.5$ and $\pi = 0.02$, and check against (a).

$\pi = 0.5$: odds 6, posterior $6/7 = 0.857$. $\pi = 0.02$: odds $0.0204 \times 6 = 0.122$, posterior 0.109. Applying (a) to $r_t = 0.857$ gives the same 0.109: the generative route just swaps the prior; the discriminative route reaches the same answer through the odds.

4.7 Problem 4.7: Confident but unfamiliar

Problem 4.7 (★★)

Two classes on one feature, equal priors: $X | (Y = 0) \sim \mathcal{N}(0, 1)$, $X | (Y = 1) \sim \mathcal{N}(4, 1)$. A reject rule accepts the predicted class c^* only if $\ln p(x | c^*) \geq \tau$. Use $\ln(2\pi) = 1.8379$; 99% of a standard normal lies within ± 2.576 .

Tools you need

Posterior from class scores; the reject option $\ln p(x | c^*) \geq \tau_{c^*}$ [one-read §4.8].

(a) Compute $\hat{p}(y = 1 | x = 9)$.

$\ln \frac{p(9|1)}{p(9|0)} = -\frac{(9-4)^2}{2} + \frac{9^2}{2} = -12.5 + 40.5 = 28$, so $\hat{p}(y = 1 | 9) = 1/(1 + e^{-28}) \approx 1 - 7 \times 10^{-13}$. Essentially certain.

(b) Choose τ so that 99% of each class's own points are accepted.

Accept when $|x - \mu_{c^*}| \leq 2.576$, i.e. $\tau = -\frac{1}{2} \ln(2\pi) - \frac{2.576^2}{2} = -0.919 - 3.318 = -4.237$.

(c) Is $x = 9$ accepted? And $x = 2$?

$x = 9$: $\ln p(9 | 1) = -0.919 - 12.5 = -13.42 < -4.237$: **rejected**, despite a posterior of almost exactly 1. The point is 5 standard deviations from class 1: the model is sure *which* class, not that it is *any* class it knows.

$x = 2$: the posterior is exactly 0.5 (equidistant, equal priors), and $\ln p(2 | c) = -0.919 - 2 = -2.919 \geq -4.237$: **accepted**, but maximally ambiguous.

Two different kinds of doubt: at $x = 2$ the classes overlap (ambiguity, seen in the posterior); at $x = 9$ the input is unfamiliar (seen only in the likelihood). A posterior alone detects the first, never the second.